

Key Management

- How to distribute key in public key set up

• DAVID
(e_D, d_D)

• ALICE
(e_A, d_A)

BOB
(e_B, d_B)

OSCAR
(e_O, d_O)

Distribution of public key -

- public announcement ✓
 - publicly available directory ✓
 - public key authority ✓
 - public key certificates ✓
- to maintain PKD
- publish this PKD in the form of book
- vulnerability

Public key directory (PKD)

Name	key
	10 10 11 10 11 0

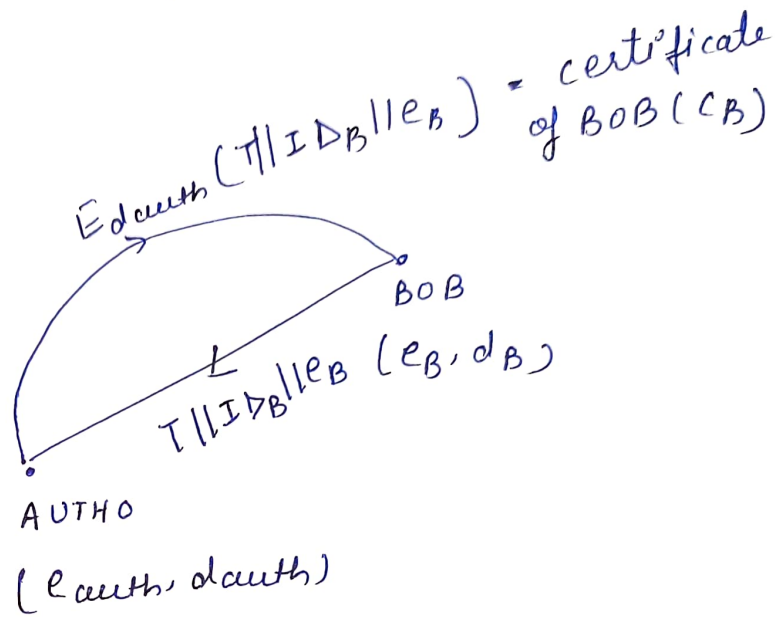
Read only files

Available in public domain (web site)

- * Attacker can also change public key for other person.
- * whole system will be disturb.

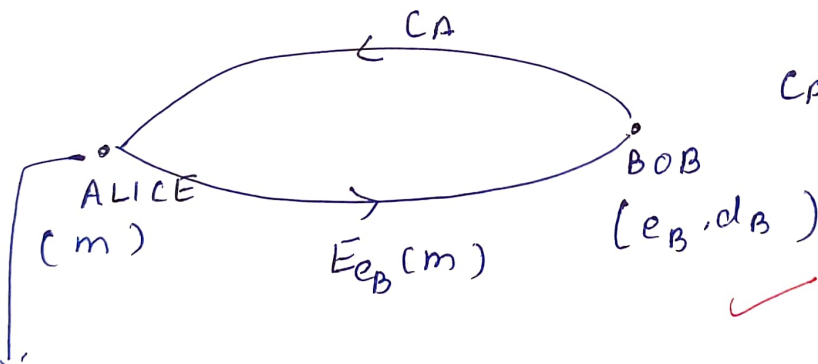
Public key certificate $\rightarrow$ (authority issue a certificate)

• ALICE



T = Timestamp
e.g.: 2 days

After certificate



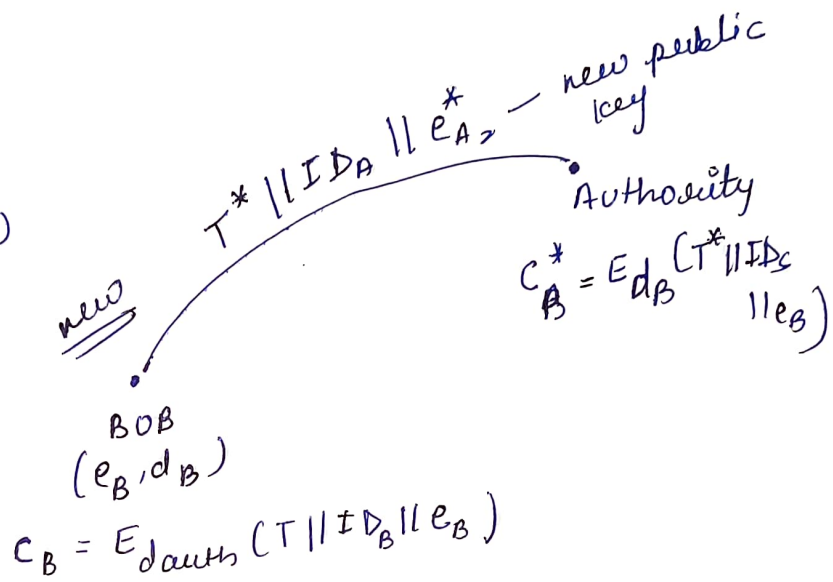
$$CA = E_{d_{auth}}(T || ID_B || e_B)$$

Decrypt
using auth public
key on CA

$$= D_{e_{auth}}(E_d(T || ID_B || e_B)) = T || ID_B || e_B$$

RSA Setup

- ① $p \neq q$
- ② $n = pq$
- ③ $ed \equiv 1 \pmod{\phi(n)}$
 $\swarrow$ $\searrow$
pub pri



Key Exchange.

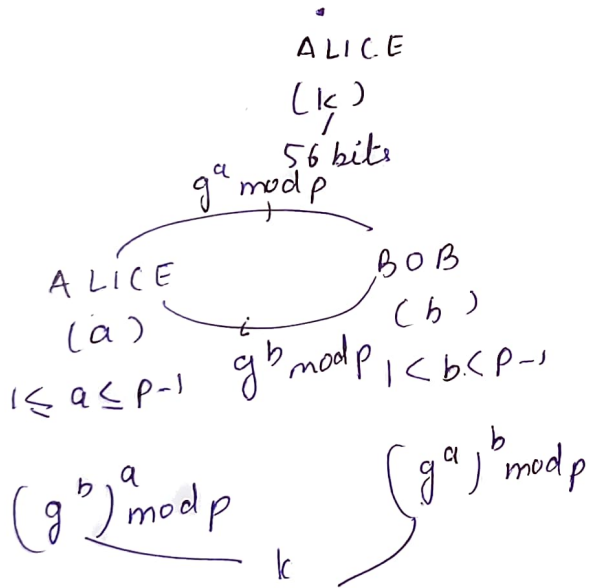
Private key (Symmetric key) common key (K)

Public key (No common key)

↓
very expensive operations

enc/Dec is usually
much faster than
public key.

Let say ALICE & BOB decided that they
are going to use DES for commⁿ
ALICE



How we
can share k

b/w ALICE & BOB
is known to be
key exchange.

One such protocol is
DHKE (Diffie Hellman
key exchange)
 $(\mathbb{Z}_p^*, *) \rightarrow$ cyclic group
 $g \rightarrow$ generator

Active attacks on DHKE
(man in the middle)

"Man in the Middle Attack"

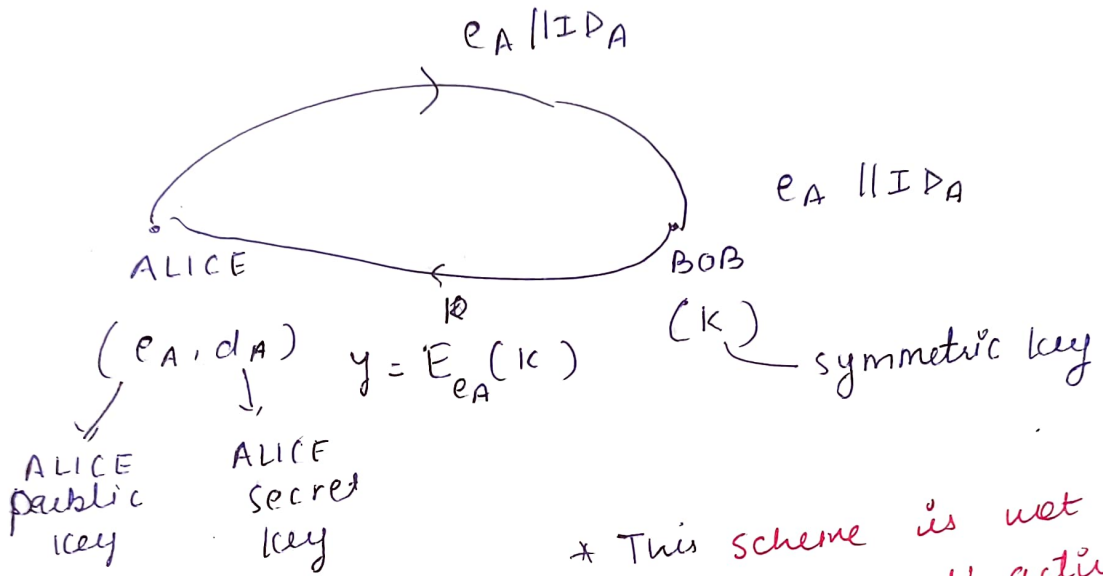
ALICE
(a)

BOB
(b)

OSCAR
(c)

Public key setup? How we can distribute secret key under (symmetric)

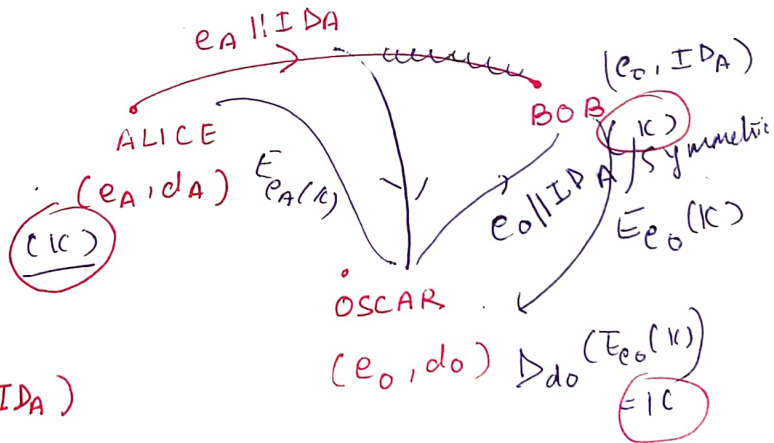
- key generation of public key is done.



* This scheme is not secure under an active attack.

RSA

Supposed to be a secure scheme



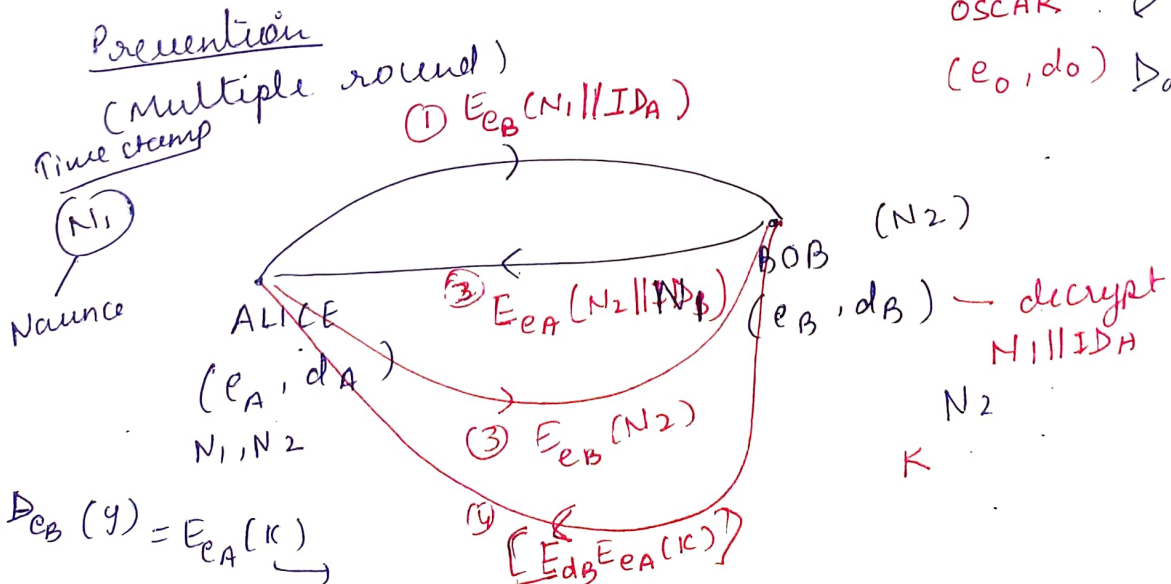
Prevention

repetition
(multiple round)
(1) E_{eff}

Time stamp

211

Naunce

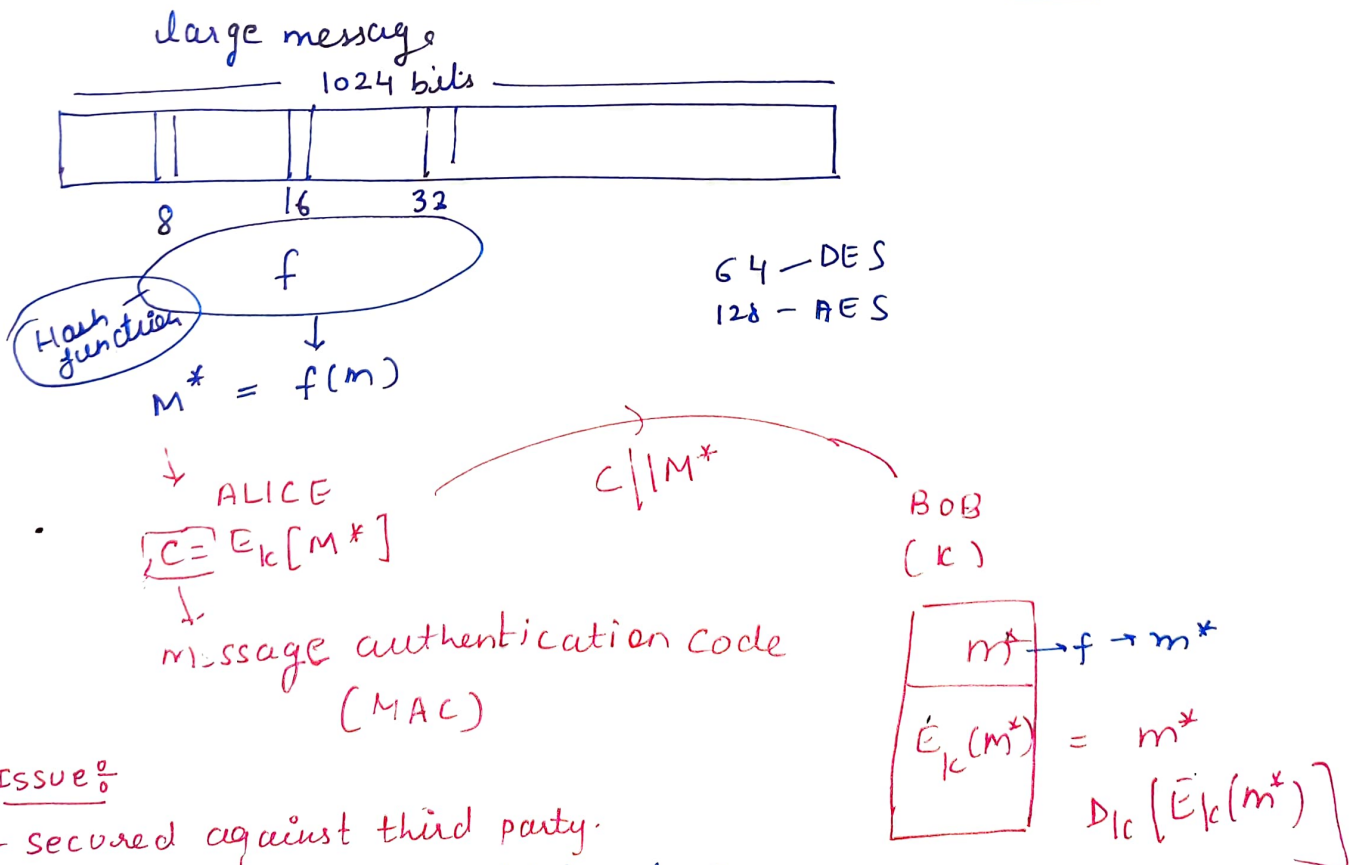
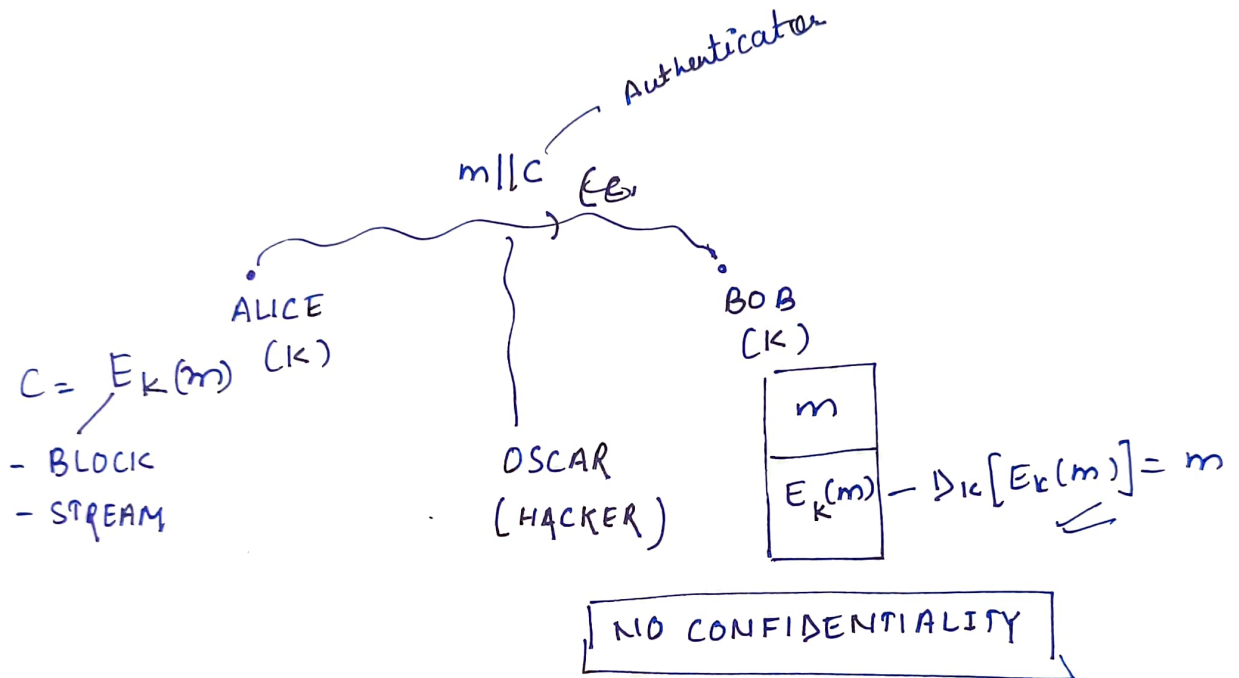


$$D_{CB}(y) = E_{CA}(K) \rightarrow$$

Message Authentication MAC

* Authentication means BOB has to ensure that message is from ALICE

① Symmetric key set up



Issue:

- 1- secured against third party.
(Bob can behave maliciously)

Problem - Non-Repudiation

② Hash Function - To solve symbol table problem

ID	→ Key(x)
Name	
CGPA	
,	
.	

store n Records in table such that able to perform dynamic operations

INSERT (T, x)

DELETE (T, x)

SEARCH (T, k)

For this purpose we need a data structure.

Let say array
key values are {4, 3, 5, 2}

Let say 825 - memory wastage

0	0	1	1	1	1
0	1	2	3	4	5

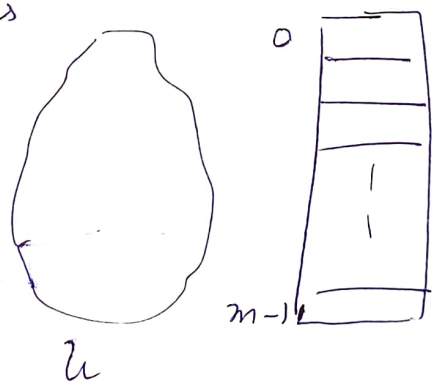
- Direct access table

* To solve this prob. hash function is used.

hash fun - $h: U \rightarrow \{0, 1, \dots, m-1\}$

$U \rightarrow$ set of all possible keys

Problem $\rightarrow$ collision
(cannot be a bijective mapping)



How to handle collision

- chaining method (linked list)

* uniform distribution of keys among the slots (good hash fun)

$n =$ number of keys

$m =$ number of slots

$$P(|U_i = j|) = \frac{1}{m}$$

load factor $\alpha = \frac{n}{m}$ (expected no. of keys in each slot)

$m = 10$
 $n = 100$
 $\alpha = \boxed{10}$

Search (T, k)

$(1 + \alpha)$ — search time

Examples of Hash function

I Division method — keys are integer ~~m~~ → prime

$$h(k) = k \bmod m$$

$$m = 2^w$$

key → 12 bits

$$m = 2^5$$

$$\begin{array}{r} 110111001011 \\ \hline \end{array} \quad \begin{array}{r} 01011 \\ \hline \end{array}$$

$k \bmod m$

high significant bits has no ~~confur~~ contribution in the hash value.

2^7 keys are colliding over a particular slot
→ not a good choice.

$m \rightarrow$ prime not close to $2^w, 10^w$

II multiplication method

$$m = 2^w$$

key — $K \rightarrow w$ bits
 $\otimes A \rightarrow w$ bits const
select

$$h(k) = (A \cdot k \bmod 2^w) \text{ right shift } (w-r) \text{ bits}$$

example

$$m = 8 \quad 2^3$$

$$w = 7$$

$$A = 1011001$$

$$K = 1101011$$

$$\begin{array}{r} A \cdot K \\ (1001010 | 0110011) \bmod 2^7 \end{array}$$

$$\boxed{0000011} \rightarrow h(k)$$

Birthday Paradox

↳ Suppose that we have group of k people
then what should be size of k such that
there is at least one common birthday
with prob. more than 0.5

$$K = 100$$

$$= .9999$$

$$n = 365 \text{ in a year}$$

$$P(n, k) \rightarrow \text{at least one duplicate}$$

$$P(n, k) = 1 - \text{no duplicate}$$

$$= 1 - \frac{\frac{n!}{n-k!}}{n^k}$$

$n(n-1)(n-2) \dots (n-k+1)$ possibilities

k all

Total no. of

$$\text{cases} = n^k$$

$$= \frac{n!}{n-k!}$$

$$1 - x < e^{-x} = 1 - \left[\left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \left(1 - \frac{(k-1)}{n}\right) \right]$$

$$P(n, k) > 1 - e^{-\frac{1}{n} - \frac{2}{n} - \frac{3}{n} - \dots - \frac{(k-1)}{n}}$$

$$> 1 - e^{-\left[\frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \dots + \frac{(k-1)}{n}\right]}$$

$$\dots > 1 - e^{-\frac{k(k-1)}{2n}}$$

$$0.5 > 1 - e^{-\frac{k(k-1)}{2n}}$$

$$e^{-\frac{k(k-1)}{2n}} > 0.5 = \frac{1}{2}$$

taking log both sides

$$\log 2 = \cancel{1 - e^{-\frac{k(k-1)}{2n}}}$$

$$\Rightarrow \frac{k^2}{2n}$$

$$k^2 \leq 2 \ln(2) \times n$$

$$k = 1.18 \sqrt{n} \Rightarrow \sqrt{n}$$

$$\Rightarrow 23$$

$$n = 365$$